



Nebiyu Waliyi Tekesa

Nationality: Ethiopian **Date of birth:** 05/05/1995 **Place of birth:** Ethiopia

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ABOUT ME

Nebiyu is currently pursuing a PhD in Sustainable Development and Climate Change at IUSS Pavia and the University of Pavia, with a research focus on extreme rainfall events in the context of climate change, particularly across Central and Northern Italy.

He holds an MSc in Climate Change and a BSc in Meteorology and Hydrology. Before his doctoral studies, Nebiyu served as a researcher and lecturer at Arba Minch University in Ethiopia from August 2021 until commencing his study leave (January 2025). Nebiyu is a member of the international climate science initiative Fresh Eyes on CMIP (Coupled Model Intercomparison Project), a scientific project organized by the World Climate Research Programme (WCRP).

His research interests include:

- Extreme Precipitation events and their hydrological risk
- Climate Modelling
- Climate change impact assessment
- Adaptation and mitigation strategies

WORK EXPERIENCE

Arba Minch University – Arba Minch, Ethiopia

Lecturer and Researcher

[10/09/2021 – 30/11/2025]

- Supervision of MSc and BSc theses.
- Conducting research in the areas of atmospheric science, climate science, climate change, and IDF curve development (stationary).
- Delivering a practical lecture on climate models and climate data downscaling.

CMIP IPO

Fresh Eye Members on CMIP: Data Analyst

[01/08/2023 – Current]

EDUCATION AND TRAINING

PhD, Sustainable Development and Climate Change

University School for Advanced Studies, IUSS Pavia [01/01/2025 – Current]

City: Pavia | **Country:** Italy | **Website:** <https://www.iusspavia.it/it>

Msc, Climate change and development

Arba Minch University [01/09/2018 – 10/08/2021]

Address: 21 Arba Minch (Ethiopia) | Website: www.amu.edu.et

- I followed courses like Advanced GIS and Remote Sensing (Remote Sensing and Image Interpretation), Atmospheric Dynamics and Numerical Weather Prediction, Tropical Weather and Climate, The Science of Climate Change, Computer Programming (Python), Earth System Modeling, Advanced Statistical Methods in Atmospheric Science, Climate Change: Impacts and Adaptation, Weather Prediction and Climate Modeling, Climate Change: Mitigation and Policies, and Agricultural and Hydrological Modeling, hydrological principles and monitoring, and surface Hydrology and others.

Bsc, Meteorology and Hydrology

Arba Minch University [01/09/2014 – 28/06/2018]

Address: 21 Arba Minch (Ethiopia) | Website: www.amu.edu.et | Final grade: 3.83/4

I followed courses like Hydrology, Meteorology science, Atmospheric dynamics, Geographic Information Systems, Remote sensing and Image processing, Satellite and Radar Meteorology, Climate change adaptation and Mitigation, Science of climate change, Introduction to computer and data processing, Computer Program with FORTRAN, Environmental hydrology, Tropical Meteorology, General Oceanography and Marine Meteorology, Applied Hydrology, Hydrological Analysis and Forecasting, Soil Physics and Vegetation, Groundwater Hydrology, Hydrogeology, Earth System Science, General Climatology, Water Resource Planning and Management, Irrigated Land Hydrology, Synoptic Meteorology, Weather Analysis, and Forecasting, Introduction to Numerical Weather Prediction, Flood and Drought, and others.

LANGUAGE SKILLS

Mother tongue(s): Afaan oromo

Other language(s):

English

LISTENING B2 READING C2 WRITING C2

SPOKEN PRODUCTION B2 SPOKEN INTERACTION B2

Amharic

LISTENING C2 READING C2 WRITING C2

SPOKEN PRODUCTION C2 SPOKEN INTERACTION C2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

meteorology / perform lectures / create solutions to problems / cooperate with colleagues / accept own accountability / follow company standards / work as a team / use geographic information systems

PROJECTS

I am one of the winners of the Research Support Insurance (RSI) by DSMT-E project, sponsored by the Bill and Melinda Gates Foundation.

CONFERENCES AND SEMINARS

Conferences

- Distinguished speaker at the 2nd Global Summit on Earth Science and Climate Change held in September 2023, in London, UK.
- Attendant of WCRP Open Science Conference 2023 held at Kigali, Rwanda.
- One of the organizers and presenters of Making the Most of CMIP Data: Access, Analysis, and Tools at AOGS2024, which was held in June 2024 in South Korea.

Short term Trainings

- Training course on Climate Data Analysis and Artificial Intelligence in the Global South” organized by the World Climate Research Programme (WCRP) and Stratosphere-Troposphere Process and Their Role in Climate (SPARC) jointly at the University of Rwanda College of Science and Technology, Kigali, Rwanda, from October 29 to 31, 2023.
- A training course on basic Python for environmental data processing, “Train the Trainers,” was held at Arba Minch University for 10 days (70 hours).
- Training course on “Climate Model Data Analysis” (using CDO, GrADS, and Python) from April 23–28, 2018, organized by Arba Minch University, Ethiopia
- A training course on Climate Data Tools (CDT), organized by Arba Minch University in collaboration with IRI, Columbia University, was held at Arba Minch University for five days.

PUBLICATIONS

Publications

- Bibi, T. S., & Tekesa, N. W. (2024). Study rainfall intensity, duration frequency relationships under climate change for the design of efficient urban stormwater drainage systems in Dodola town, Ethiopia. *International Journal of Hydrology Science and Technology*, 17(2), 220-238.
- Bibi, T. S., & Tekesa, N. W. (2023). Impacts of climate change on IDF curves for urban stormwater management systems design: the case of Dodola Town, Ethiopia. *Environmental Monitoring and Assessment*, 195(1), 170.
- Dingley, B., Li, Y., Rao, Y., Tekesa, N. W., Galytska, E., Prasad, A., FALL, C. M. N., Katezenberger, A., Palmeiro, F., & Tolhurst, G. (2024, June 27). Making the most of CMIP data: access, analysis, and tools. Zenodo. <https://doi.org/10.5281/zenodo.12558016>.

COMMUNICATION AND INTERPERSONAL SKILLS

INTERPERSONAL & DIGITAL SKILLS

- I am working with international scholars, and I can easily interact with people from abroad, with cultural and social backgrounds.
- I presented scientific matters and research outputs at different local and international events. As a result, I have experience making oral presentations and am very interested in public communication science.
- Climate Modelling, Hydrological Modelling, Statistical analysis, Python, R, MATLAB

REFERENCES

References

- Professor Gabriella Petaccia, Professor in Hydraulic Constructions and Hydrology, Department of Civil and Architectural Engineering, University of Pavia, Pavia, Italy. gabriella.petaccia@unipv.it, She is my main supervisor for my Phd project.
- Dr. Thomas Torora Minda, Assistant Professor (Ph.D.) in the faculty of meteorology and hydrology, Arba Minch University, Ethiopia, thomas.torora@amu.edu.et, (+251) 931410359. He was the supervisor of my MSc thesis and course lecturer.
- Dr. Tadesse Tujuba Kenea, Associate Professor (Ph.D.) in the faculty of meteorology and hydrology, Arba Minch University, Ethiopia, tade2j@gmail.com, (+25 1) 946692840. He is the dean of the faculty of meteorology and hydrology at the Water Technology Institute, Arba Minch University and he was my lecturer.